IT PROJECT DOCUMENTATION
VERSION 1.0
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Revision History

Version	Date	Author	Description
1.0	2025-07-02	Sylvain Simo	Initial Draft

Project Overview

PROJECT NAME: DNS SERVER SETUP USING VAGRANT

Purpose:

The purpose of this project is to provision and configure a DNS server in a virtualized environment using Vagrant and VirtualBox. This allows testing DNS configurations in a contained and reproducible development environment.

Scope:

This project involves creating a Vagrantfile to provision four Ubuntu VMs (1 DNS server and 3 clients), installing BIND9 on the DNS server, and configuring zone files and resolver settings via Bash scripts.

Objectives:

- Automate provisioning of multiple VMs using a Vagrantfile
- Configure one VM as a DNS server using a Bash script
- Configure client VMs to use the DNS server for name resolution
- Create and test a DNS zone for the domain **digitallumens123.net**

Stakeholders:

- Project Lead: Sylvain Simo
- End Users: Developers and IT Infrastructure Teams

Requirements

Functional Requirements:

- **Vagrantfile** provisions four Ubuntu VMs
- **DNS service (BIND9)** must be installed and configured on one VM
- Client VMs must resolve domain via DNS server

Non-Functional Requirements:

- VM provisioning must be automated and repeatable
- DNS resolution should complete in <100ms within LAN

Business Requirements:

- Ability to test DNS configurations in an isolated network
- Scalable base for extending into hybrid cloud simulation

System Architecture

Architecture Diagram:
(Host → Vagrant → 4 VMs: 1 DNS Server, 3 Clients)

Description:
The host machine runs a Vagrant-managed multi-VM environment. One VM is designated as the DNS server running BIND9. The remaining three VMs are configured as DNS clients using static resolver settings. Two Bash scripts automate configuration: one for DNS server setup, another for client setup.

Technology Stack

Layer	Technology
Provisioning	Vagrant + VirtualBox
OS	Ubuntu 22.04
DNS Server	BIND9
Scripting	Bash

Implementation Plan

Milestones:

1. Write and validate Vagrantfile for 4 VMs

```
VAGRANTFILE_API_VERSION = "2"

Vagrant.configure(VAGRANTFILE_API_VERSION) do |config|
    config.vm.box = "ubuntu/jammy64"
    config.vm.synced_folder ".", "/vagrant", disabled: true

    # Base settings for all VMs
    config.vm.provider "virtualbox" do |vb|
      vb.memory = 512
      vb.cpus = 1
    end

    machines = {
      "ns-service" => "192.168.6.110",
      "web"        => "192.168.6.111",
      "mail"       => "192.168.6.112",
      "nfs"        => "192.168.6.113"
    }

    machines.each do |name, ip|
      config.vm.define name do |node|
        node.vm.hostname = "#{name}.digitallumens123.net"
        node.vm.network "private_network", ip: ip
        node.vm.provision "shell", path: (name == "ns-service" ? "scripts/dns-setup.sh" : "scripts/client-setup.sh")
      end
    end
end
```

2. Create dns-server-config.sh to install and configure BIND9

```bash
#!/bin/bash

echo "Installing BIND9..."
apt update
apt install -y bind9 bind9utils bind9-doc

echo "Setting up zone files..."

mkdir -p /etc/bind/zones

cat > /etc/bind/named.conf.local <<EOF
zone "digitallumens123.net" {
    type master;
    file "/etc/bind/zones/db.digitallumens123.net";
};

zone "6.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/zones/db.192.168.6";
};
EOF

# Forward zone
cat > /etc/bind/zones/db.digitallumens123.net <<EOF
\$TTL 86400
@   IN  SOA ns-service.digitallumens123.net. admin.digitallumens123.net. (
    2025070600 ; Serial
    3600       ; Refresh
    1800       ; Retry
    604800     ; Expire
    86400 )    ; Minimum TTL

@       IN  NS      ns-service.digitallumens123.net.
ns-service  IN  A   192.168.6.110
web         IN  A   192.168.6.111
mail        IN  A   192.168.6.112
nfs         IN  A   192.168.6.113
chat        IN  CNAME mail.digitallumens123.net.

@           IN  MX  10 mail.digitallumens123.net.
EOF

# Reverse zone
cat > /etc/bind/zones/db.192.168.6 <<EOF
\$TTL 86400
@   IN  SOA ns-service.digitallumens123.net. admin.digitallumens123.net. (
    2025070600 ; Serial
    3600       ; Refresh
    1800       ; Retry
    604800     ; Expire
    86400 )    ; Minimum TTL

@   IN  NS  ns-service.digitallumens123.net.
10      IN  PTR ns-service.digitallumens123.net.
11      IN  PTR web.digitallumens123.net.
12      IN  PTR mail.digitallumens123.net.
13      IN  PTR nfs.digitallumens123.net.
EOF

systemctl restart bind9
```

3. Create dns-client-config.sh to update resolv.conf on clients

```
#!/bin/bash

# Point clients to the DNS server (ns-service)
echo "nameserver 192.168.6.110" > /etc/resolv.conf
```

4. Test resolution using **dig** or **nslookup**

```
vagrant@mail:~$ dig @192.168.6.110 mail.digitallumens123.net

; <<>> DiG 9.18.30-0ubuntu0.22.04.2-Ubuntu <<>> @192.168.6.110 mail.digitallumens123.net
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 29897
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 8db952eba6b8dc30010000006864cefb30f773d48e183e4a (good)
;; QUESTION SECTION:
;mail.digitallumens123.net.     IN      A

;; ANSWER SECTION:
mail.digitallumens123.net. 86400 IN     A       192.168.6.112

;; Query time: 28 msec
;; SERVER: 192.168.6.110#53(192.168.6.110) (UDP)
;; WHEN: Wed Jul 02 06:17:31 UTC 2025
;; MSG SIZE  rcvd: 98
```

Deliverables:

- Vagrantfile
- dns-server-config.sh
- dns-client-config.sh
- DNS configuration files (named.conf.local, db.digitallumens123.net)

Directory Structure:

```
dns-lab-vagrant/
├── Vagrantfile
├── scripts/
|   ├── dns-setup.sh
|   ├── client-setup.sh
```

Timeline:

- Estimated Time: 1 Day

Testing Plan

Test Strategy:
Manual testing using dig or nslookup on all client machines

Test Cases:

1. dig digitallumens123.net from client returns configured IP
2. ping digitallumens123.net from client is successful
3. DNS server responds within acceptable latency

Results Summary:
Initial test showed NXDOMAIN; resolved by correcting zone file and configuration in BIND9

Deployment

Deployment Steps:

1. Install Vagrant (Install | Vagrant | HashiCorp Developer) and Virtualbox (Downloads – Oracle VirtualBox) on the host computer
2. From Command prompt as Admin, Create and Organize project folders structure according to the above directory structure
3. Create Vagrantfile and the 2 bash script files according t the above scripts
4. Make scripts executable:

   chmod +x scripts/*.sh

5. Run **vagrant up** to provision all four VMs and their configurations from the scripts
6. SSH into each client VM to test the result: **vagrant ssh clienthostname**
7. Test name resolution from client VMs:

   dig web.digitallumens123.net

   dig chat.digitallumens123.net

dig -x 192.168.6.112

Rollback Plan:

- Use **vagrant destroy -f** to reset environment
- Edit scripts and re-provision as needed

Access Credentials:

- Vagrant default (vagrant/vagrant)

Maintenance and Support

Monitoring Tools:

- journalctl and systemctl for BIND9

Logging:

- DNS logs stored in /var/log/syslog

Support Contacts:

- Sylvain Simo (Admin)

Appendices

Additional Resources:

- BIND9 Documentation: https://bind9.readthedocs.io
- Vagrant Docs: https://developer.hashicorp.com/vagrant/docs

Glossary:

- BIND: Berkeley Internet Name Domain
- DNS: Domain Name System
- VM: Virtual Machine

References:

- Project notes from 2025-07-01 DNS setup session
- dig output logs for digitallumens123.net
- dns-server-config.sh and dns-client-config.sh contents